

SOUJANYA PONNAPALLI

soujanya@berkeley.edu · soujanyaponnapalli.github.io · [scholar profile](#)

WORK & EDUCATION	University of California, Berkeley Postdoc · Sky Computing Lab · EECS Department Supervisors: Prof. Natacha Crooks and Prof. Matei Zaharia	UC Berkeley 2023-Ongoing
	University of Texas at Austin PhD · Systems and Storage Lab · CS Department Advisor: Prof. Vijay Chidambaram Minimizing I/O Bottlenecks to Achieve Scalable and High-Throughput Systems	UT Austin 2017-23
	International Institute of Information Technology, Hyderabad Bachelor's with Honors · SERC Lab · CS and Engineering Advisor: Prof. Suresh Purini	IIIT-H 2013-17
INTERESTS	Distributed systems · Storage systems · Operating systems · AI systems I am a systems researcher. I build systems that are not just fast and scalable but also reliable, providing strong guarantees such as performance isolation and fault tolerance in practice.	

†Lead postdoc *Project co-leads

PUBLICATIONS	1. Metronome: I/O-Efficient Logging for State Machine Replication [VLDB'27] Harald Ng*, Soujanya Ponnappalli *†, Kevin Harrison, Reginald Frank, Audrey Cheng, Natacha Crooks, and Paris Carbone Under revision · Preprint
	2. Delta Fair Sharing: Performance Isolation for Multi-Tenant Storage Systems [VLDB'27] Tyler Griggs*, Soujanya Ponnappalli *†, Dev Bali, Wenjie Ma, James DeLoye, Audrey Cheng, Jaewan Hong, Natacha Crooks, Scott Shenker, Ion Stoica, and Matei Zaharia Under submission · Preprint
	3. Time is Here for Just-in-Time Systems: Challenges and Opportunities [MLForSys'26] Shu Liu, Alexander Krentsel, Shubham Agarwal, Mert Cemri, Ziming Mao, Soujanya Ponnappalli , Alexandros G. Dimakis, Sylvia Ratnasamy, Matei Zaharia, Aditya G. Parameswaran, and Ion Stoica Under submission · Preprint
	4. Rollbaccine: Herd Immunity against Storage Rollback Attacks in TEEs [SIGMOD'26] David C. Y. Chu, Aditya Balasubramanian, Dee Bao, Natacha Crooks, Heidi Howard, Lucky Kathanas, and Soujanya Ponnappalli †
	5. Supporting Our AI Overlords: Redesigning Data Systems to be Agent-First [CIDR'26] Shu Liu, Soujanya Ponnappalli , Shreya Shankar, Sepanta Zeighami, Alan Zhu, Shubham Agarwal, Ruiqi Chen, Samion Suwito, Shuo Yuan, Ion Stoica, Matei Zaharia, Alvin Cheung, Natacha Crooks, Joseph E. Gonzalez, and Aditya G. Parameswaran
	6. SkyStore: Cost-Optimized Object Storage Across Regions and Clouds [VLDB'25] Shu Liu, Xiangxi Mo, Moshik Hershcovitch, Henric Zhang, Audrey Cheng, Guy Girmonsky, Gil Vernik, Michael Factor, Tiemo Bang, Soujanya Ponnappalli †, Natacha Crooks, Joseph E. Gonzalez, Danny Harnik, and Ion Stoica [Best of VLDB'25 nomination]

PUBLICATIONS

7. **Real Life Is Uncertain. Consensus Should Be Too!** [HotOS'25]
Reginald Frank*, Octavio Lomeli, Neil Giridharan, **Soujanya Ponnappalli***[†], Marcos K. Aguilera, and Natacha Crooks
8. **Lost in Translation: The Search for Meaning in Network-Attached AI Accelerator Disaggregation** [HotNets'25]
Jaewan Hong, Yifan Qiao, **Soujanya Ponnappalli**, Shu Liu, Marcos K. Aguilera, Vincent Liu, Christopher J. Rossbach, and Ion Stoica
9. **DINOMO: An Elastic, Scalable, High-Performance Key-Value Store for Disaggregated Persistent Memory** [VLDB'22]
Sekwon Lee, **Soujanya Ponnappalli**, Sharad Singhal, Marcos K. Aguilera, Kimberly Keeton, and Vijay Chidambaram
10. **RainBlock: Faster Transaction Processing in Public Blockchains** [ATC'21]
Soujanya Ponnappalli, Aashaka Shah, Souvik Banerjee, Dahlia Malkhi, Amy Tai, Vijay Chidambaram, and Michael Wei
11. **WineFS: Hupage-Aware File System for PM that Ages Gracefully** [SOSP'21]
Rohan Kadekodi, Saurabh Kadekodi, **Soujanya Ponnappalli**, Harshad Shirwadkar, Gregory R. Ganger, Aasheesh Kolli, and Vijay Chidambaram
12. **Software-Defined Data Protection: Low Overhead Policy Compliance at the Storage Layer is Within Reach!** [VLDB'21]
Zsolt István, **Soujanya Ponnappalli**, and Vijay Chidambaram
13. **Finding Crash-Consistency Bugs with Bounded Black-Box Crash Testing** [OSDI'18]
Jayashree Mohan, Ashlie Martinez, **Soujanya Ponnappalli**, Pandian Raju, and Vijay Chidambaram
14. **mLSM: Making Authenticated Storage Faster in Ethereum** [HotStorage'18]
Pandian Raju, **Soujanya Ponnappalli**, Evan Kaminsky, Gilad Oved, Zachary Keener, Vijay Chidambaram, and Ittai Abraham
15. **Holographic Storage for the Cloud: Advances and Challenges** [TOS'25]
Nathanaël Cherié, Jiaqi Chu, Grace Brennan, Pashmina Cameron, Pedro Da Costa, Jannes Gladrow, Guilherme Ilunga, Douglas Kelly, Sarah Lewis, Joowon Lim, Giorgio Maltese, Tony Mason, Greg O'Shea, **Soujanya Ponnappalli**, Michael Rudow, Alan Sanders, Theano Stavrinou, Xingbo Wu, Mengyang Yang, Dushyanth Narayanan, Benn Thomsen, and Antony Rowstron
16. **CrashMonkey and ACE: Systematically Testing File-System Crash Consistency** [TOS'19]
Jayashree Mohan, Ashlie Martinez, **Soujanya Ponnappalli**, Pandian Raju, and Vijay Chidambaram

[†]Lead postdoc

PREPRINTS

17. **Token Latency Fairness: Performance Isolation for Multi-Tenant LLM Serving** [arXiv'26]
Dev Bali, **Soujanya Ponnappalli**[†], Yichuan Wang, Natacha Crooks, Scott Shenker, and Matei Zaharia
18. **SKYE: Write-Optimized Key-Value Store with Fine-Grained Control over Persistent Memory Accesses** [arXiv'26]
Soujanya Ponnappalli, Sekwon Lee, Rohan Kadekodi, and Vijay Chidambaram

PATENTS

- Recoverable Processes** [US Patent App. 17/981,296]
Jonathan D. Goldstein, Philip A. Bernstein, **Soujanya Ponnappalli**, Jose M. Faleiro, and Peter Charles Shrosbree

		†Lead postdoc *Project co-leads
ONGOING PROJECTS	Page Cache Fairness: Performance Isolation for Multiple Tenants in Operating Systems Soujanya Ponnappalli [†] , Charisse Ivana Yeung, Natacha Crooks, and Matei Zaharia	
	Powder: Fine-Grained Fault-Tolerance for Distributed Storage Systems Reginald Frank*, Soujanya Ponnappalli* [†] , Neil Giridharan, Naama Ben-David, and Natacha Crooks	
	Fugue: Exploring Durability Trade-Offs in Replicated Storage Diogo Antunes, Baltasar Dinis, Soujanya Ponnappalli, Vijay Chidambaram, Peter Druschel, and Rodrigo Rodrigues	
WORK EXPERIENCE	Microsoft Research, Redmond Mentor: Jonathan Goldstein	Summer'22
	Microsoft Research, Redmond Mentor: Anirudh Badam	Summer'20
	Microsoft Research, Cambridge Mentors: Dushyanth Narayanan and Antony Rowstron	Summer'19
	VMware Research, California Mentors: Michael Wei and Dahlia Malkhi	Summer'18
TEACHING EXPERIENCE	Teaching Assistant at UT Austin Virtualization with Prof. Vijay Chidambaram	Fall'20,'23
	Teaching Assistant at IIIT-H Algorithms and Data Structures with Prof. Kishore Kothapalli Operating Systems with Prof. Suresh Purini Electrical Science with Prof. Rambabu Kalla	2015-17
SERVICE	NSF Proposal Reviewer	2026
	Program Committee	OSDI'26, NSDI'26, ATC'25, NSDI'25, EuroSys'25
	External Program Committee	FAST'25, ATC'24, NSDI'19
	Journal Reviewer, ACM TOCS	2024
	Hallway Discussion Lead, SOSR	2021
	Shadow PC, EuroSys	2020
	Chair, Graduate Application Assistance Program (GAAP@UT)	2020-21
MENTORING	PhD students	
	Reginald Frank, Shu Liu, Tyler Griggs, Dev Bali, and Diogo Antunes	UC Berkeley
	David C. Y. Chu and Jaewan Hong	UC Berkeley
	Harald Ng	KTH Sweden
	Masters students	
	James DeLoye, Extending Delayed Fair Sharing	UC Berkeley
	Undergraduate students	
	Charisse Ivana Yeung	UC Berkeley
	Souvik Banerjee, Implementing Stateless Clients in Ethereum	UT Austin
	Gilad Oved, Evan Kaminsky, and Zachary Keener	UT Austin

LEADERSHIP	Organizer, Sky Systems Seminar, Sky Computing Lab, UC Berkeley	2024-25
	Co-organizer, Databases Seminar, Sky Computing Lab, UC Berkeley	2024-25
	Representative, Graduate Association of Computer Sciences (GRACS@UT)	2020-21
	Mentor, Women in Computer Science (WiCS), UT Austin	2019-20
	Co-organizer, Systems Seminar, Lab for Advanced Systems Research, UT Austin	2018
AWARDS & GRANTS	Summer Research Institute (SuRI) Fellowship, EPFL	2025
	Berkeley RDI Frontier Research Funding	2024
	The James C. Browne Graduate Fellowship	2017-18
	IIIT-H Best All-Rounder Gold Medalist	2017
	Dean's Award for ranking in the top 5% of students, IIIT-H	2014-17
REFERENCES	Natacha Crooks and Matei Zaharia (UC Berkeley), Vijay Chidambaram (UT Austin), Marcos K. Aguilera (NVIDIA), and Tianyin Xu (UIUC)	